A Design Method of Ultra-wideband High Efficiency Power Amplifier

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***Abstract*—This paper presents a novel method for designing an ultra-wideband dual-mode power amplifier. The proposed method involves a comprehensive analysis of the principles underlying existing resistive-reactive Class F and Class J power amplifiers. It examines the conditions necessary to achieve high efficiency and subsequently improves upon the existing design methods for resistive-reactive Class F and Class J amplifiers by expanding their bandwidth. The ultimate goal is to develop a design methodology that enables the realization of high-efficiency power amplifiers over a broader frequency range. Utilizing CG2H40010F GaN HEMT transistors, a dual-mode broadband high-efficiency power amplifier has been designed and fabricated. This power amplifier operates within the frequency band from 1.9 GHz to 2.9 GHz. Measurements indicate that the drain efficiency ranges from 65% to 70.5%, the gain exceeds 12 dB, and the output power is greater than 40 dBm. The designed power amplifier achieves an efficiency of over 65% across a 1 GHz bandwidth, thereby demonstrating the feasibility and effectiveness of the proposed design method.**

Keywords—Ultra-wideband; High efficiency; Power amplifier

Introduction

With the rapid development of communication technologies, the need to enhance the bandwidth and efficiency of radio frequency (RF) power amplifiers (PAs) has become increasingly critical. In applications such as 5G/6G communications, radar systems, and electronic warfare, the exponential growth in data transmission demands imposes stricter requirements on the bandwidth and efficiency of RF PAs. For instance, 6G networks may operate in the terahertz (THz) frequency range, necessitating RF components with higher operating frequencies and broader bandwidths.

Moreover, signal transmission in high-frequency bands suffers from significant path loss, requiring higher transmit power. High-efficiency power amplifiers can deliver sufficient output power while minimizing energy consumption. Consequently, expanding bandwidth and improving efficiency are essential for the advancement of RF power amplifiers, playing a pivotal role in enhancing communication system performance and meeting future technological requirements.

This paper proposes a novel design methodology for a dual-mode ultra-wideband power amplifier (PA). Based on an analysis of the operational principles and high-efficiency conditions of resistive-reactance Class-F and Class-J PAs, we introduce an output matching network technique capable of simultaneously controlling harmonic impedance and fundamental matching across a broad frequency range. By employing mode-combination theory, this approach extends the operational bandwidth of such amplifiers, ultimately enabling the realization of a dual-mode broadband high-efficiency PA design.

I. Theoretical Analysis

A. Fundamental Principles of Resistive-Reactance Class-J Power Amplifiers

The operational modes of conventional continuous-mode power amplifiers (PAs) [1-2]are realized by multiplying the voltage expression of high-efficiency PAs by a reactance factor $(1-\gamma\sin\theta)(1-\gamma\sin\theta)$, where γ is a tuning parameter. The introduction of an imaginary component serves to expand the harmonic matching region, transforming the harmonic impedance from a single point into a trajectory along the periphery of the Smith chart. This expansion increases the feasible harmonic matching region for high-efficiency PA modes, thereby relaxing the constraints on harmonic control and enhancing the operational bandwidth of the amplifier. However, in practical implementations, achieving perfect matching of a purely imaginary second-harmonic impedance while simultaneously satisfying broadband fundamental matching is nearly impossible.

To address this limitation, V. Carrubba et al. [3] proposed augmenting the reactance factor $(1-\gamma\sin\theta)(1-\gamma\sin\theta)$ with a resistive factor $(1+\delta\cos\theta)(1+\delta\cos\theta)$, where δ is a tuning parameter. This resistive extension relaxes the second-harmonic impedance condition from a purely imaginary value to a complex impedance located within the Smith chart, thereby further expanding the feasible harmonic matching region.

Figure 1 illustrates the impedance requirements for continuous Class-F power amplifiers(CFPA), while Figure 2 presents the corresponding impedance conditions for resistive-reactance Class-F PAs.(RRFPA)

A comparative analysis of these figures reveals that the resistive-reactance Class-F configuration provides a significantly expanded impedance matching space compared to its continuous Class-F counterpart. This enhanced design space facilitates more relaxed harmonic control requirements, thereby enabling the realization of broader operational bandwidth in resistive-reactance Class-F power amplifiers.

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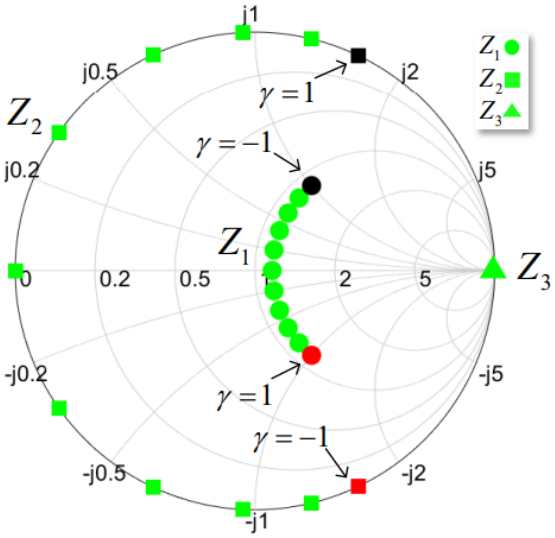


Fig. 1. the impedance design space for CCFPA

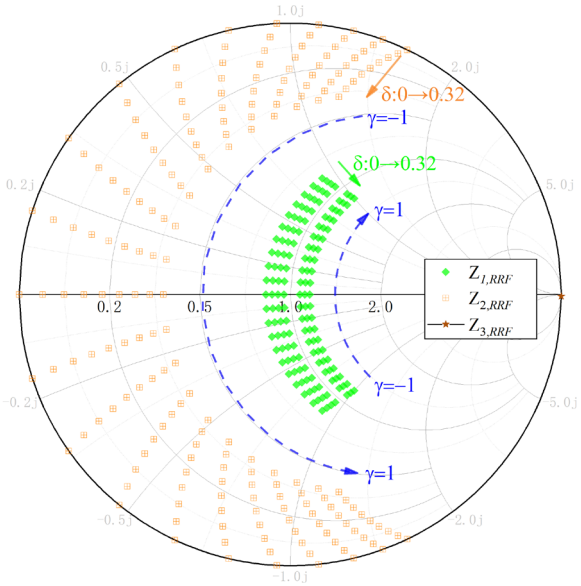


Fig. 2. the impedance design space of RRCFPA

B. *Fundamental Principles of Resistive-Reactance Class-F Power Amplifiers*

Similarly, the voltage expression of the resistive-reactance Class-F power amplifier is derived by multiplying the conventional continuous Class-F voltage expression with the resistive factor (1 + δ cosθ). When δ = 0, the theoretical maximum output efficiency of this amplifier class reaches 90.6%. However, when δ ≠ 0, both the fundamental power and drain efficiency experience degradation.

Figures 3 and 4 respectively depict the impedance design spaces for conventional Class-J(CJPA) and resistive-reactance Class-J power amplifiers(RRCJPA), demonstrating the expanded design flexibility achieved through resistive-reactance modification. Figure 5 further examines the fundamental and harmonic impedance characteristics of both resistive-reactance Class-J and Class-F configurations as functions of tuning parameters γ and δ. The parameter γ is swept across [-1, 1] for both amplifier classes, while the parameter δ operates within [0, 0.17] for resistive-reactance Class-J and [0, 0.32] for resistive-reactance Class-F modes. Notably, increasing δ produces two significant effects: (1) the second-harmonic impedance region expands inward from the Smith chart periphery, and (2) the fundamental impedance shifts outward from the chart center. This dual impedance transformation mechanism underlies the enhanced bandwidth capabilities of resistive-reactance architectures compared to their conventional counterparts.

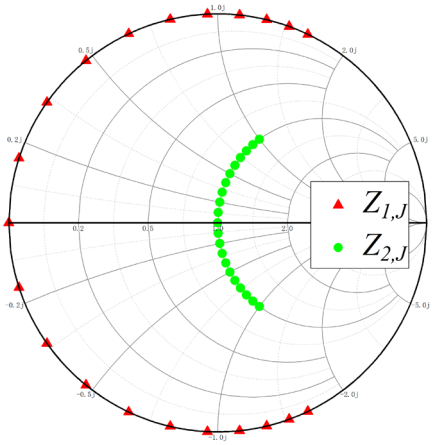


Fig. 3. the impedance design space for CJPA

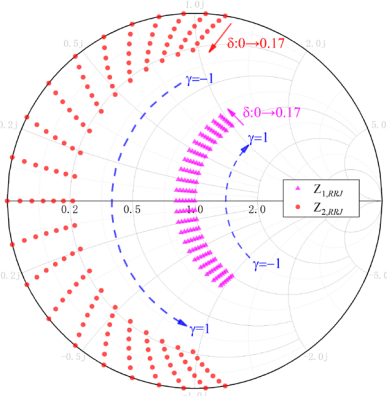


Fig. 4. impedance design space of RRCJPA

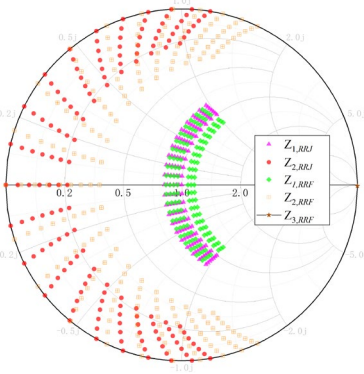


Fig. 5. impedance design spaces for both RRCJPA and CCFPA

The frequency response characteristics of LC networks exhibit a clockwise rotation with increasing frequency. As previously discussed, both amplifier types - particularly the resistive-reactance Class-F configuration - demonstrate

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harmonic impedance conditions that similarly follow this clockwise progression. Through analysis of LC network frequency responses and the required harmonic impedance positions for continuous-mode operation on the Smith chart, we establish that strategic clockwise selection and combination of harmonic impedance regions enables the hybridization of multiple continuous-mode PA architectures.

Building upon these analytical findings, this work demonstrates that targeted clockwise combination of specific harmonic impedances permits the realization of dual-mode hybrid continuous-class PAs. Specifically, we propose a novel resistive-reactance Class-F/J dual-mode hybrid continuous-class power amplifier architecture. This design extends beyond conventional single-mode high-efficiency amplifiers by simultaneously maintaining high efficiency while significantly enhancing operational bandwidth.

C. Matching Network Design for Resistive-Reactance Continuous-Mode Power Amplifiers

The selected active device for this power amplifier is the CG2H40010F GaN HEMT (Gallium Nitride High-Electron-Mobility Transistor) from CREE Inc., which exhibits high gain, excellent thermal stability, and superior high-voltage robustness across the designed frequency band. Furthermore, the package parasitic equivalent network of this transistor has been characterized and documented in prior literature, providing a validated reference for precise impedance matching and stability analysis.

Traditional Class-F, Class-J, and Class-F^{-1} power amplifiers typically employ multiple λ/4 transmission lines for harmonic impedance control[4-6]. This work innovatively integrates the harmonic control module with the fundamental matching module by utilizing the stopband suppression characteristics of low-pass filters, simultaneously achieving fundamental impedance matching and harmonic impedance control. This approach significantly simplifies the output network while minimizing the amplifier's physical dimensions.

Continuous-mode power amplifiers derive their impedance configurations from theoretical current-source models. However, in practical implementations using packaged transistors like the CG2H40010F, the matching network must perform impedance transformation from the input port to the package interface, effectively converting the impedance at the transistor's output plane to that required at the current-source plane. The package equivalent circuit exhibits capacitive characteristics at high frequencies, which can be modeled as equivalent capacitance, while the Chebyshev low-pass filter prototype demonstrates inductive characteristics in its stopband, equivalent to inductance. This combined pathway forms an LC parallel resonant circuit that generates significant impedance at specific resonant frequencies. According to resonance theory, parallel structures produce high impedance at their resonant frequencies, thereby creating favorable conditions for third-harmonic control in continuous Class-F operation.

The low-pass broadband network can generate the required frequency-dependent impedances at the current-source plane of the CG2H40010F's drain across a wide bandwidth. The tabular method proposed by Matthaei G L [7] provides an efficient and straightforward approach to implementing specific impedance transformations in low-pass circuits. Building upon the research of Fano R M et al. 8-11 and considering the target bandwidth of this design, a sixth-order Chebyshev low-pass filter was ultimately selected for the output matching network implementation.

As the operational frequency band of power amplifiers increases, signal transmission becomes dominated by electromagnetic field propagation confined between conductors and reference ground planes. This high-frequency behavior results in non-uniform voltage and current distributions along the circuit paths, rendering traditional lumped-parameter networks inadequate for accurate implementation. Consequently, the design methodology requires conversion of the synthesized lumped elements into distributed components prior to final realization.

II. MEASUREMENT

The corresponding impedance plot in Figure 6 demonstrates that the fundamental frequencies are tightly clustered around the center of the Smith chart, while harmonic components exhibit well-controlled behavior - with 4.2 GHz impedances approaching the short-circuit point and 6.3 GHz impedances near the open-circuit region. These results confirm that at the design center frequency of 2.1 GHz, the circuit simultaneously satisfies near-ideal short-circuit conditions for the second harmonic (4.2 GHz) and open-circuit conditions for the third harmonic (6.3 GHz), while maintaining optimal fundamental matching.

This configuration meets both classical Class-F harmonic requirements at 2.1 GHz and enables resistive-reactance continuous Class-F operation across the adjacent frequency bands from 1.5-2.9 GHz. The successful control of both fundamental and harmonic impedances through a unified matching network demonstrates significant advantages in circuit simplification while maintaining broadband performance and multi-mode operation capability.

The impedance convergence observed in Figures 7-8 validates the effectiveness of the Chebyshev low-pass prototype approach for broadband matching and harmonic control, particularly in its ability to handle the package-to-current-source plane transformation. This methodology achieves a remarkable 58% fractional bandwidth (1.5-2.9 GHz) while maintaining excellent matching characteristics and proper harmonic terminations throughout the operational band.

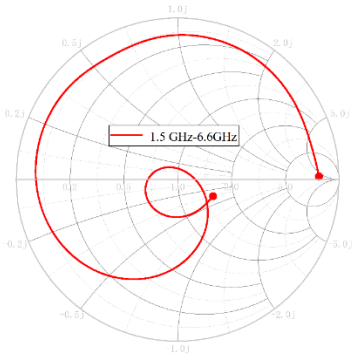


Fig. 6. impedance trajectory at the current-source plane on the Smith chart

The fabricated power amplifier prototype is shown in Figure 7, with compact dimensions of 99 mm × 41 mm. The circuit is implemented on Rogers 4350B substrate (thickness H = 0.762 mm, relative permittivity ϵ_r = 3.66).

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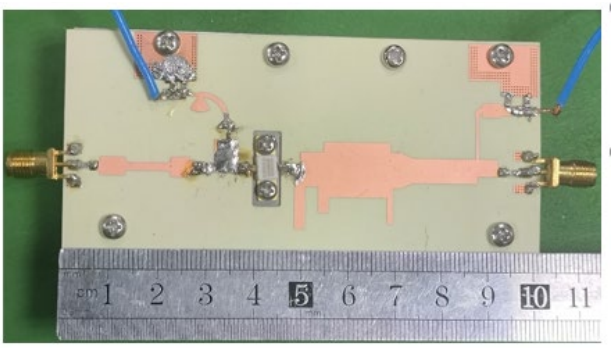


Fig. 7. fabricated power amplifier prototype

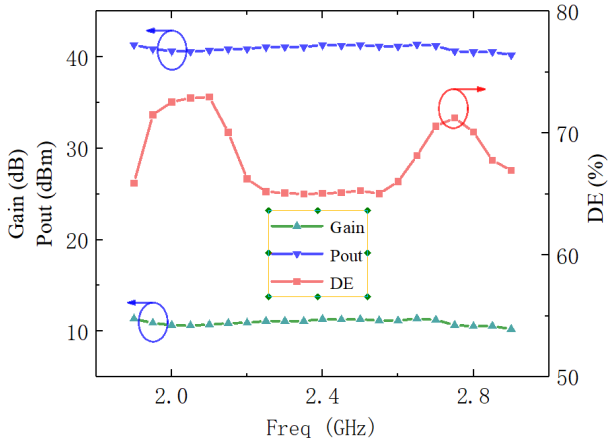


Fig. 8. the large-signal simulation results of the designed power amplifier

Figure 8 presents the large-signal simulation results of the designed power amplifier, demonstrating excellent performance across the 1.9-2.9 GHz bandwidth. The measured saturated output power ranges from 40.1 to 42 dBm, with a power gain of 12.1-12 dB and drain efficiency maintained between 65-70.5%. These results confirm the effectiveness of the proposed design methodology, as the amplifier achieves consistently high drain efficiency exceeding 65% throughout the entire operational bandwidth while maintaining outstanding power and gain characteristics. The performance metrics validate the successful implementation of the harmonic impedance control technique and unified matching network architecture.

III. Summary

This paper proposes an ultra-wideband high-efficiency power amplifier (PA) design methodology. By analyzing the harmonic impedance conditions of resistive-reactance Class-J and Class-F PAs, the approach combines these two high-efficiency modes to achieve bandwidth extension. Furthermore, the design utilizes the stopband rejection characteristics of low-pass filters to replace traditional λ/4 transmission lines for harmonic impedance control while simultaneously achieving fundamental impedance matching, thereby simplifying the PA output matching network. Based on this ultra-wideband design approach, a dual-mode hybrid ultra-wideband high-efficiency power amplifier has been successfully implemented. Measurement results demonstrate that the prototype achieves 65-70.5% drain efficiency, 40.1-42 dBm output power, and 12.1-12 dB gain across the 1.9-2.9 GHz frequency band. These experimental results validate the effectiveness of the proposed ultra-wideband PA design methodology.

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